Model Derivations

Companion notes to "Trade Tariffs and Exchange Rates"

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These notes derive three models: the two-country benchmark, the two-layer nested three-country model with the reversal threshold, and the canonical competing-exporters model from the preferential-trade-agreement (PTA) literature. Prose is limited to defining variables and linking equations.

1 Two-country model

1.1 Setup

Countries $i \in \{A, B\}$. Goods: tradable T_i (produced only in i, consumed in both) and non-tradable N_i. Labor is the only factor, immobile across countries, endowment L_i.

Technology (linear):

$$Y_{T_i} = A_{T_i} L_{T_i}, \qquad Y_{N_i} = A_{N_i} L_{N_i}, \qquad L_{T_i} + L_{N_i} = L_i. \tag{1}$$

Perfect competition and the normalization $P_{T_i} = 1$ (producer price of each tradable in its own currency) give

$$w_i = A_{T_i}, \qquad P_{N_i} = \frac{w_i}{A_{N_i}} = \frac{A_{T_i}}{A_{N_i}}. \tag{2}$$

Let e be the nominal exchange rate: units of A's currency per unit of B's currency. The price of T_B in A's currency is e; the price of T_A in B's currency is $1/e$.

Preferences (Cobb–Douglas, common weights):

$$U_i = C_{T_A i}^{\alpha_{T_A}} \, C_{T_B i}^{\alpha_{T_B}} \, C_{N_i}^{\alpha_N}, \qquad \alpha_{T_A} + \alpha_{T_B} + \alpha_N = 1, \tag{3}$$

where C_{gi} is country i's consumption of good g.

1.2 Demands

Cobb–Douglas expenditure shares are constant. With income I_i and consumer prices $P_g^{(i)}$ (in i's currency),

$$P_g^{(i)} C_{gi} = \alpha_g I_i \quad \Longrightarrow \quad C_{gi} = \frac{\alpha_g I_i}{P_g^{(i)}}. \tag{4}$$

Under free trade $I_i = w_i L_i$, $P_{T_B}^{(A)} = e$, $P_{T_A}^{(B)} = 1/e$.

1.3 Equilibrium under free trade

Country A's trade balance in A's currency is export revenue minus import spending. From (4), B's purchases of T_A are $C_{T_A B} = \alpha_{T_A} I_B/(1/e) = \alpha_{T_A} w_B L_B\, e$, valued at producer price 1 in A's currency, and A's purchases of T_B are $C_{T_B A} = \alpha_{T_B} w_A L_A/e$, valued at price e. Therefore

$$\mathrm{TB}_A = 1 \cdot C_{T_A B} - e\, C_{T_B A} = \alpha_{T_A} w_B L_B\, e - \alpha_{T_B} w_A L_A. \tag{5}$$

Setting $\mathrm{TB}_A = 0$ (by Walras' law this is the only independent condition):

$$\boxed{e = \frac{\alpha_{T_B}}{\alpha_{T_A}} \cdot \frac{w_A L_A}{w_B L_B} = \frac{\alpha_{T_B}}{\alpha_{T_A}} \cdot \frac{A_{T_A} L_A}{A_{T_B} L_B}} \tag{6}$$

1.4 Tariff

Country A levies ad valorem tariff $\tau \geq 0$ on imports of T_B. The consumer price in A becomes $(1+\tau)e$; producers in B still receive 1. Revenue is rebated lump-sum:

$$I_A = w_A L_A + T_A^{\mathrm{rev}}, \qquad T_A^{\mathrm{rev}} = \tau e\, C_{T_B A}. \tag{7}$$

Using $C_{T_B A} = \alpha_{T_B} I_A/[(1+\tau)e]$,

$$I_A = w_A L_A + \frac{\tau}{1+\tau}\,\alpha_{T_B} I_A \quad\Longrightarrow\quad I_A = \frac{w_A L_A}{1 - \dfrac{\alpha_{T_B}\tau}{1+\tau}}. \tag{8}$$

The trade balance (flows valued at producer prices; the tariff wedge is a domestic transfer):

$$\mathrm{TB}_A = C_{T_A B} - e\, C_{T_B A} = \alpha_{T_A} w_B L_B\, e \;-\; \frac{\alpha_{T_B} I_A}{1+\tau}. \tag{9}$$

Setting $\mathrm{TB}_A = 0$ and substituting (8):

$$\boxed{e(\tau) = \frac{1}{1+\tau} \cdot \frac{1}{1 - \dfrac{\alpha_{T_B}\tau}{1+\tau}} \cdot \frac{\alpha_{T_B}}{\alpha_{T_A}} \cdot \frac{A_{T_A} L_A}{A_{T_B} L_B}} \tag{10}$$

Result 1.1. *$e(\tau)$ is strictly decreasing in τ for $\alpha_{T_B} \in (0,1)$: a tariff appreciates the tariff-imposing country's currency.*

Proof. Write $e(\tau) = e(0) \cdot f(\tau)$ with $f(\tau) = \big[(1+\tau) - \alpha_{T_B}\tau\big]^{-1} = \big[1 + (1-\alpha_{T_B})\tau\big]^{-1}$. Since $1 - \alpha_{T_B} > 0$, $f' < 0$. $\square$

1.5 Relative wage, terms of trade, and real exchange rate

Relative wage. Wages in own currency are $w_A = A_{T_A}$ and $w_B = A_{T_B}$, both constant. The relative wage in a common currency (A's) is

$$\omega \;\equiv\; \frac{e\,w_B}{w_A} \;=\; \frac{A_{T_B}}{A_{T_A}}\,e \;\propto\; e. \tag{11}$$

A rise in e (depreciation of A's currency) is a fall in A's wage relative to B's.

Terms of trade. A's terms of trade are the price of its exports relative to its imports, in a common currency and at world (pre-tariff) prices:

$$\mathrm{ToT}_A \;\equiv\; \frac{P_{T_A}}{e\,P_{T_B}} \;=\; \frac{1}{e}. \tag{12}$$

Combining (11) and (12): the exchange rate, the (inverse) common-currency relative wage, and the (inverse) terms of trade are the same object up to constants,

$$e \;\propto\; \frac{w_B\,e}{w_A} \;\propto\; \frac{1}{\mathrm{ToT}_A}. \tag{13}$$

This holds because producer prices are unit labor costs ($P_{T_i} = w_i/A_{T_i} = 1$): one factor and constant returns make the identification exact.

Real exchange rate. The consumer price index in each country is the Cobb–Douglas index over consumer prices (constants κ_i collect the share terms):

$$P_A = \kappa_A\left[P_{T_A}\right]^{\alpha_{T_A}}\left[(1+\tau)e\right]^{\alpha_{T_B}}\left[P_{N_A}\right]^{\alpha_N} = \kappa_A\left[(1+\tau)e\right]^{\alpha_{T_B}}\left(\tfrac{A_{T_A}}{A_{N_A}}\right)^{\alpha_N}, \tag{14}$$

$$P_B = \kappa_B\left[1/e\right]^{\alpha_{T_A}}\left[P_{T_B}\right]^{\alpha_{T_B}}\left[P_{N_B}\right]^{\alpha_N} = \kappa_B\,e^{-\alpha_{T_A}}\left(\tfrac{A_{T_B}}{A_{N_B}}\right)^{\alpha_N}. \tag{15}$$

The real exchange rate is

$$q \;\equiv\; \frac{e\,P_B}{P_A} \;=\; \mathrm{const}\times e^{\,1-\alpha_{T_A}-\alpha_{T_B}}\,(1+\tau)^{-\alpha_{T_B}} \;=\; \mathrm{const}\times e^{\,\alpha_N}\,(1+\tau)^{-\alpha_{T_B}}, \tag{16}$$

so in logs

$$d\log q \;=\; \alpha_N\,d\log e \;-\; \alpha_{T_B}\,\frac{d\tau}{1+\tau}. \tag{17}$$

Equation (17) has three implications. First, q moves with e, attenuated by the non-tradable share α_N. Second, the tariff appreciates A in real terms directly, at any e, because it raises A's consumer price index. Third, since $P_{N_i} = A_{T_i}/A_{N_i}$ is a productivity ratio, non-tradables affect the level of q but enter its response to trade shocks only through the exponent α_N.

1.6 The Marshall–Lerner condition

Let export volume depend on the price foreign buyers face, $p^* = 1/e$, and import volume on the price home buyers face, $p = (1+\tau)e$. Write $X(p^*)$ and $M(p)$ with elasticities

$$\varepsilon_X \equiv -\frac{X'(p^*)\,p^*}{X} \;\geq\; 0, \qquad \varepsilon_M \equiv -\frac{M'(p)\,p}{M} \;\geq\; 0. \tag{18}$$

The trade balance in A's currency, at producer prices, is

$$\mathrm{TB}_A(e,\tau) = X(1/e) \;-\; e\,M\big((1+\tau)e\big). \tag{19}$$

Response to the exchange rate. Differentiate (19) with respect to e at $\tau = 0$:

$$\frac{\partial \mathrm{TB}_A}{\partial e} = -\frac{X'(1/e)}{e^2} - M - e\,M'(e) = \frac{\varepsilon_X X}{e} - M + \varepsilon_M M, \tag{20}$$

using $X' = -\varepsilon_X X e$ and $eM' = -\varepsilon_M M$. Evaluated at balanced trade ($X = eM$):

$$\boxed{\frac{\partial \mathrm{TB}_A}{\partial e} = M\big(\varepsilon_X + \varepsilon_M - 1\big)} \tag{21}$$

A depreciation improves the trade balance if and only if $\varepsilon_X + \varepsilon_M > 1$. This is the Marshall–Lerner condition. The term -1 in (21) is the valuation effect: a depreciation raises the home-currency cost of each unit of imports, which worsens the balance one-for-one. The two volume responses must offset this effect.

Response to the tariff. Differentiate (19) with respect to τ at fixed e:

$$\frac{\partial \mathrm{TB}_A}{\partial \tau} = -e\,M'\big((1+\tau)e\big)\,e = \frac{\varepsilon_M}{1+\tau}\,eM \;>\; 0 \quad \text{whenever } \varepsilon_M > 0. \tag{22}$$

A tariff improves the trade balance at a fixed exchange rate as long as import demand slopes down. (With a revenue rebate, income effects modify (22); in the Cobb–Douglas model of Section 1.4 the producer-price import value is $\alpha_{T_B} I_A/(1+\tau) \propto [1 + (1-\alpha_{T_B})\tau]^{-1}$, still decreasing in τ.)

Why Marshall–Lerner is necessary and sufficient for the conventional wisdom. Equilibrium requires $\mathrm{TB}_A(e(\tau),\tau) = 0$. By the implicit function theorem,

$$\frac{de}{d\tau} = -\frac{\partial \mathrm{TB}_A/\partial \tau}{\partial \mathrm{TB}_A/\partial e}. \tag{23}$$

The numerator is positive by (22): the tariff creates a surplus at the initial exchange rate. To restore balance, the exchange rate must move in the direction that worsens the trade balance. If Marshall–Lerner holds, that direction is appreciation, $de < 0$, which is the conventional result. If Marshall–Lerner fails, depreciation worsens the balance instead, and the same tariff-induced surplus requires $de > 0$, so the conventional result reverses. The sign of $de/d\tau$ is therefore the opposite of the sign of $\varepsilon_X + \varepsilon_M - 1$. This is why the condition is both necessary and sufficient.

The Cobb–Douglas case. With constant expenditure shares, import volume is $M = \alpha_{T_B} I/p$, so $\varepsilon_M = 1$; similarly $\varepsilon_X = 1$. Then $\varepsilon_X + \varepsilon_M = 2 > 1$ and (21) is positive. Marshall–Lerner holds automatically, which is why the closed form (10) is monotone in τ.

Multilateral remark. In the three-country model of Section 2, each bilateral relationship satisfies a Marshall–Lerner condition, and the tariff still improves A's overall balance at fixed exchange rates. The bilateral conditions do not determine how the required adjustment is split between e_{AB} and e_{AC}. That split depends on substitution across foreign origins, governed by ρ, relative to substitution between home and foreign goods, governed by η. This is the subject of Section 2.

2 Three-country nested model

2.1 Setup

Countries $i \in \{A, B, C\}$, each producing tradable T_i and non-tradable N_i with the same linear technology, wage equation, and normalization as (2). Exchange rates e_{ij} (i's currency per unit of j's), $e_{ii} = 1$, no-arbitrage $e_{ij} = e_{ik}e_{kj}$; the free rates are e_{AB}, e_{AC}. Tariffs $\tau_{ij} \geq 0$ (i's tariff on j's good). Consumer price of T_j in i: $p_{ij} = (1 + \tau_{ij})\, e_{ij}$.

Preferences (two-layer nest):

$$C_i = C_{T_i}^{\alpha_T}\, C_{N_i}^{1-\alpha_T} \qquad \text{(Cobb–Douglas outer)} \tag{24}$$

$$C_{T_i} = \left[\alpha_D^{1/\eta}\, C_{T_i i}^{\frac{\eta-1}{\eta}} + (1-\alpha_D)^{1/\eta}\, M_i^{\frac{\eta-1}{\eta}} \right]^{\frac{\eta}{\eta-1}} \qquad \text{(home vs. imports, elasticity } \eta\text{)} \tag{25}$$

$$M_i = \left[\textstyle\sum_{j\neq i} \beta_j^{1/\rho}\, C_{T_j i}^{\frac{\rho-1}{\rho}} \right]^{\frac{\rho}{\rho-1}} \qquad \text{(across origins, elasticity } \rho\text{)} \tag{26}$$

with home weight α_D, origin weights β_j (symmetric case: $\beta_j = 1/2$). The single-elasticity model is the special case $\rho = \eta$.

2.2 Price indices and expenditure shares

CES price indices (in i's currency):

$$P_{M_i} = \left[\textstyle\sum_{j\neq i} \beta_j\, p_{ij}^{1-\rho} \right]^{\frac{1}{1-\rho}}, \qquad P_{T_i} = \left[\alpha_D\, p_{ii}^{1-\eta} + (1-\alpha_D)\, P_{M_i}^{1-\eta} \right]^{\frac{1}{1-\eta}}. \tag{27}$$

Expenditure shares of total income I_i (rows sum to α_T):

$$s_{ii} = \alpha_T\, \alpha_D \left(\frac{p_{ii}}{P_{T_i}} \right)^{1-\eta}, \tag{28}$$

$$s_{ij} = \alpha_T\, (1-\alpha_D) \left(\frac{P_{M_i}}{P_{T_i}} \right)^{1-\eta} \beta_j \left(\frac{p_{ij}}{P_{M_i}} \right)^{1-\rho}, \qquad j \neq i. \tag{29}$$

2.3 Income and trade balances

Tariff revenue is rebated lump-sum. A fraction $\tau_{ij}/(1+\tau_{ij})$ of spending $s_{ij}I_i$ is revenue, so

$$I_i = w_i L_i + \sum_{j\neq i} \frac{\tau_{ij}}{1+\tau_{ij}}\, s_{ij} I_i \quad\implies\quad I_i = \frac{w_i L_i}{1 - \sum_{j\neq i} \frac{\tau_{ij}}{1+\tau_{ij}}\, s_{ij}}. \tag{30}$$

The shares depend only on prices, so (30) is a closed form.

Trade balances at producer prices (spending $s_{ij}I_i$ is tariff-inclusive; divide by $1+\tau_{ij}$ to get the producer-price value):

$$\mathrm{TB}_i = \sum_{k\neq i} \frac{s_{ki}\, I_k}{1+\tau_{ki}} - \sum_{j\neq i} \frac{s_{ij}\, I_i}{1+\tau_{ij}}, \tag{31}$$

with all quantities expressed in a common currency (this does not affect the zeros). Equilibrium: $\mathrm{TB}_B = \mathrm{TB}_C = 0$; $\mathrm{TB}_A = 0$ follows by Walras' law.

2.4 Linearization at the symmetric point

Set $L_i = 1$, $A_{T_i} = A_{N_i} = 1$, $\beta_j = 1/2$, and evaluate at free trade ($\tau_{ij} = 0$), where $e_{AB} = e_{AC} = 1$ and all incomes equal 1. Baseline shares:

$$s_{ii} = \alpha_T \alpha_D \equiv a, \qquad s_{ij} = \tfrac{1}{2}\,\alpha_T(1-\alpha_D) \equiv \tfrac{m}{2}, \qquad m \equiv \alpha_T(1-\alpha_D). \tag{32}$$

Perturb by $d\tau$ (A's tariff on B) and let $\nu_j \equiv d\log e_{Aj}$ ($\nu_A = 0$). Log price changes: $d\log p_{ij} = \nu_j + \mathbf{1}\{(i,j) = (A,B)\}\, d\tau$.

Index changes, from (27):

$$d\log P_{M_i} = \sum_{j\neq i} \beta_j \, d\log p_{ij}, \qquad d\log P_{T_i} = \alpha_D \, d\log p_{ii} + (1-\alpha_D)\, d\log P_{M_i}. \tag{33}$$

Explicitly:

$$d\log P_{M_A} = \tfrac{1}{2}(\nu_B + \nu_C + d\tau), \qquad d\log P_{M_B} = \tfrac{1}{2}\nu_C, \qquad d\log P_{M_C} = \tfrac{1}{2}\nu_B. \tag{34}$$

Share changes, from (28)–(29):

$$d\log s_{ii} = (1-\eta)\big(d\log p_{ii} - d\log P_{T_i}\big), \tag{35}$$
$$d\log s_{ij} = (1-\rho)\big(d\log p_{ij} - d\log P_{M_i}\big) + (1-\eta)\big(d\log P_{M_i} - d\log P_{T_i}\big). \tag{36}$$

Income changes, from (30) (the revenue term is first order for country A only):

$$d\log I_A = \tfrac{m}{2}\, d\tau, \qquad d\log I_B = \nu_B, \qquad d\log I_C = \nu_C. \tag{37}$$

Differentiate (31) for $i = B, C$ at the baseline. Each term $s\, I/(1+\tau)$ contributes $\frac{m}{2}\,(d\log s +$

$d\log I - d\tau_{\cdot}$), where $d\tau_{\cdot} = d\tau$ appears only in the A-to-B flow:

$$d\,\mathrm{TB}_B = \tfrac{m}{2}\Big[\big(d\log s_{AB} + \tfrac{m}{2}d\tau - d\tau\big) + \big(d\log s_{CB} + \nu_C\big) - \big(d\log s_{BA} + \nu_B\big) - \big(d\log s_{BC} + \nu_B\big)\Big] = 0, \tag{38}$$

$$d\,\mathrm{TB}_C = \tfrac{m}{2}\Big[\big(d\log s_{AC} + \tfrac{m}{2}d\tau\big) + \big(d\log s_{BC} + \nu_B\big) - \big(d\log s_{CA} + \nu_C\big) - \big(d\log s_{CB} + \nu_C\big)\Big] = 0. \tag{39}$$

Substituting (35)–(36) and collecting terms in $(\nu_B, \nu_C, d\tau)$ gives a linear system. Define

$$D \equiv 3\alpha_D(\eta - 1) + \rho + 1, \qquad \rho^* \equiv 3\big[\,1 + \alpha_D(\eta-1) - \alpha_T(1-\alpha_D)\,\big]. \tag{40}$$

Collecting terms yields the linear system

$$-\frac{mD}{4}\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}\begin{pmatrix} \nu_B \\ \nu_C \end{pmatrix} = \frac{m}{4}\begin{pmatrix} \rho + \rho^*/3 \\ -\rho + \rho^*/3 \end{pmatrix} d\tau. \tag{41}$$

Invert using $\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}^{-1} = \frac{1}{3}\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$:

$$\begin{pmatrix} \nu_B \\ \nu_C \end{pmatrix} = -\frac{1}{3D}\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}\begin{pmatrix} \rho + \rho^*/3 \\ -\rho + \rho^*/3 \end{pmatrix} d\tau = \frac{1}{3D}\begin{pmatrix} -(\rho+\rho^*) \\ \rho - \rho^* \end{pmatrix} d\tau. \tag{42}$$

Result 2.1 (Threshold)**.** *At the symmetric point,*

$$\left.\frac{d\log e_{AB}}{d\tau}\right|_{\tau=0} = -\frac{\rho+\rho^*}{3D} < 0, \qquad \boxed{\left.\frac{d\log e_{AC}}{d\tau}\right|_{\tau=0} = \frac{\rho-\rho^*}{3D}} \tag{43}$$

with $D > 0$ for $\eta \geq 1$ (local stability). A appreciates against B for all parameters; A depreciates against the untariffed C if and only if $\rho > \rho^*$.

2.5 Impossibility under a single elasticity

Impose $\rho = \eta = \sigma$ (the single-elasticity CES). Then

$$\rho^* - \sigma = 3 + 3\alpha_D(\sigma-1) - 3\alpha_T(1-\alpha_D) - \sigma = \sigma(3\alpha_D - 1) + 3(1-\alpha_D)(1-\alpha_T). \tag{44}$$

Result 2.2 (Impossibility)**.** *If $\alpha_D \geq 1/3$, both terms in* (44) *are nonnegative and the second is positive, so $\rho^* > \sigma$ for every $\sigma > 0$: the conventional appreciation against C survives at any elasticity. At $\alpha_D = 1/3$, $\alpha_T = 3/4$ (the symmetric baseline calibration $\alpha_{T_j} = \alpha_N = 1/4$): $\rho^* - \sigma = 1/2$ and $D = 2\sigma$, so*

$$\left.\frac{d\log e_{AC}}{d\tau}\right|_{\tau=0} = \frac{\sigma - \rho^*}{3D} = -\frac{1}{12\sigma} \;\longrightarrow\; 0^-. \tag{45}$$

2.6 Nominal versus real exchange rate in the three-country model

The quantities of Section 1.5 carry over: e_{Aj} is proportional to j's relative wage in A's currency and to the inverse of A's bilateral terms of trade with j. The real rate against C is $q_{AC} = e_{AC} P_C / P_A$ with $P_i = \kappa_i \, (P_{T_i}^{\text{own}})^{\alpha_T} (P_{N_i})^{1-\alpha_T}$, where $P_{T_i}^{\text{own}}$ is the tariff-inclusive tradable index (27) in i's own currency and $P_{N_i} = A_{T_i}/A_{N_i}$ is constant. Hence

$$d \log q_{AC} = d \log e_{AC} + \alpha_T \big(d \log P_{T_C}^{\text{own}} - d \log P_{T_A}^{\text{own}} \big). \tag{46}$$

At the symmetric point, converting the indices to own currency ($d \log P_{T_C}^{\text{own}} = d \log P_{T_C} - \nu_C$):

$$d \log P_{T_A}^{\text{own}} = (1 - \alpha_D) \, \tfrac{1}{2} (\nu_B + \nu_C + d\tau), \tag{47}$$

$$d \log P_{T_C}^{\text{own}} = \alpha_D \nu_C + (1 - \alpha_D) \tfrac{1}{2} \nu_B - \nu_C = (1 - \alpha_D) \big(\tfrac{1}{2} \nu_B - \nu_C \big). \tag{48}$$

Substituting and using $m = \alpha_T (1 - \alpha_D)$:

$$d \log q_{AC} = \nu_C + m \Big(\tfrac{1}{2} \nu_B - \nu_C - \tfrac{1}{2} \nu_B - \tfrac{1}{2} \nu_C - \tfrac{1}{2} d\tau \Big) = \Big(1 - \tfrac{3m}{2} \Big) \nu_C \; - \; \tfrac{m}{2} \, d\tau. \tag{49}$$

As in (17), the tariff appreciates A in real terms directly through the price index (the $-\frac{m}{2} d\tau$ term), and the pass-through from the nominal rate is attenuated. Setting (49) to zero with $\nu_C = \frac{\rho - \rho^*}{3D} \, d\tau$ gives the real-rate reversal threshold: $(2 - 3m)(\rho - \rho^*) = 3mD$, and since $D = 3\alpha_D (\eta - 1) + \rho + 1$ this is linear in ρ, with solution

$$\boxed{\rho_q^* = \frac{(2 - 3m) \, \rho^* + 3m \big[3\alpha_D (\eta - 1) + 1 \big]}{2 - 6m}} \qquad (\rho_q^* > \rho^* \text{ for } m \in (0, \tfrac{1}{3})). \tag{50}$$

At the benchmark ($\alpha_D = 0.8$, $\alpha_T = 0.4$, $\eta = 1.5$, so $m = 0.08$): $\rho^* = 3.96$ and $\rho_q^* = 4.93$. Real depreciation against C requires more diversion than nominal depreciation, because part of the nominal movement is absorbed by the tariff's direct effect on A's price index.

3 The N-country case

3.1 General linear structure

Countries $i \in \{1, \ldots, N\}$, country 1 the numeraire, with the same technology and nested preferences as Section 2: home share α_{D_i}, origin weights β_{ij}, elasticities (η, ρ). Let $\nu = (\nu_2, \ldots, \nu_N)$ collect the log exchange rates and let a tariff $d\tau$ be imposed by country 1 on country 2. Linearizing the $N - 1$ independent trade-balance conditions at a free-trade baseline gives

$$J \, d\nu = -F \, d\tau \quad \Longrightarrow \quad \frac{d\nu}{d\tau} = (-J)^{-1} F = \frac{\operatorname{adj}(-J) \, F}{\det(-J)}, \tag{51}$$

where $J_{ij} = \partial \mathrm{TB}_i / \partial \nu_j$ and $F_i = \partial \mathrm{TB}_i / \partial \tau$ at fixed exchange rates (the impact effects). The entries of J and F are linear in η and ρ, with coefficients given by baseline expenditure shares and incomes, exactly as in the three-country linearization of Section 2.4. Equation (51) generalizes the two-country formula (23), which is the case of a single equation: one J, one F.

Stability. The adjustment process $\dot{\nu}_i \propto \mathrm{TB}_i$ is locally stable if and only if all eigenvalues of J have negative real parts. Any such matrix satisfies $\det(-J) > 0$, so at a stable equilibrium the common denominator in (51) is positive and the sign of each response equals the sign of its numerator. This condition is the N-country analogue of the Marshall–Lerner condition: at $N = 2$ it is the scalar $\partial \mathrm{TB}_A/\partial e > 0$, that is $\varepsilon_X + \varepsilon_M > 1$ from (21); at the symmetric three-country point it is $D > 0$ from Section 2.5.

The multilateral Marshall–Lerner condition. Suppose the demand system satisfies gross substitutability. Then J has negative diagonal entries, nonnegative off-diagonal entries, and a dominant diagonal (from homogeneity and Walras' law, with the numeraire absorbing the residual). The diagonal entries are the bilateral Marshall–Lerner conditions of Section 1.6, one per country: own appreciation worsens own balance. The off-diagonal sign is the multilateral addition, with no two-country counterpart: another country's appreciation improves your balance. A matrix with this sign pattern is a stable Metzler matrix, and $-J$ is then an M-matrix. M-matrices are inverse-positive:

$$(-J)^{-1} \;\geq\; 0 \quad \text{elementwise.} \tag{52}$$

At the symmetric three-country point all of these conditions collapse into the single scalar D: the Jacobian there is $-\frac{mD}{4}\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$, so the two diagonal conditions and stability all hold if and only if $D > 0$.

Result 3.1 (Generic sign rule)**.** *At a stable equilibrium satisfying the multilateral Marshall–Lerner condition,*

$$\operatorname{sign}\!\left(\frac{d\nu_i}{d\tau}\right) = \operatorname{sign}\!\Big(\sum_j w_{ij} F_j\Big), \qquad w_{ij} \equiv \big[(-J)^{-1}\big]_{ij} > 0. \tag{53}$$

In particular, if all impact effects share a sign, all responses share it. Sign ambiguity arises only when the impact vector is mixed.

The rule contains the earlier cases. At $N = 2$ there is one weight and one impact: the impact is positive by (22), so the sign is unconditional and the currency appreciates — the conventional result of Section 1. At $N = 3$ the weights are $\frac{1}{3}\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$, and the bystander's response mixes its own impact with the target's negative impact; the threshold ρ^* of Section 2 is the value of ρ at which this weighted sum crosses zero.

Because F is linear in ρ and $\operatorname{adj}(-J)$ is a polynomial of degree $N-2$ in ρ, each numerator in (51) is a polynomial of degree $N-1$ in ρ. At $N = 3$ this is a quadratic with one relevant root, which yields the scalar threshold of Section 2. At $N \geq 4$ single crossing is not guaranteed: there exist asymmetric configurations in which the sign of a bystander's response changes twice as ρ rises, so no scalar threshold exists in general. The sign of any particular response remains computable from (53).

3.2 The symmetric case: a closed form

Let all countries have equal size, common preferences, home share α_D, and equal origin weights $1/(N-1)$. Country 1 (A) tariffs country 2 (B); the remaining $k = N-2$ bystanders are identical, with representative c. By symmetry the system (51) reduces to two unknowns, (ν_B, ν_C). Define

$$\rho_1^* \equiv 1 + \alpha_D(\eta - 1) - \alpha_T(1-\alpha_D), \qquad D_N \equiv N\alpha_D(\eta-1) + (N-2)\rho + 1. \tag{54}$$

Solving the two-equation system (the derivation parallels Section 2.4 and is verified symbolically in the replication package) gives

$$\left.\frac{d\log e_{AB}}{d\tau}\right|_{\tau=0} = -\frac{N\rho_1^* + (N-2)\rho}{N\,D_N} < 0, \tag{55}$$

$$\left.\frac{d\log e_{AC}}{d\tau}\right|_{\tau=0} = \frac{\rho - N\rho_1^*}{N\,D_N}, \tag{56}$$

$$\left.\frac{d\log e_{BC}}{d\tau}\right|_{\tau=0} = \frac{(N-1)\,\rho}{N\,D_N} > 0, \tag{57}$$

with $D_N > 0$ the stability condition; at $N=3$, $D_3 = 3\alpha_D(\eta-1) + \rho + 1 = D$. Hence:

Result 3.2 (N-country threshold)**.** *In the symmetric N-country model, an isolated tariff depreciates A's currency against each untariffed bystander if and only if*

$$\boxed{\rho > \rho^*(N) = N\big[1 + \alpha_D(\eta-1) - \alpha_T(1-\alpha_D)\big]} \tag{58}$$

At $N=3$ this is the threshold of Section 2: the coefficient 3 there is the number of countries. The threshold grows linearly in N because, with symmetric weights, each bystander receives a share $1/(N-1)$ of the diverted expenditure, so the substitutability required for any single bystander to gain scales with the number of countries.

For the symmetric trade war, add the responses to $d\tau_{AB}$ and its mirror $d\tau_{BA}$. Relabeling countries gives $\nu_C^{\text{war}} = 2\nu_C - \nu_B$, and substituting (55)–(56):

$$\left.\frac{d\log e_{AC}}{d\tau}\right|_{\tau=0}^{\text{war}} = \frac{\rho - \rho_1^*}{D_N}, \qquad \left.\frac{d\log e_{AB}}{d\tau}\right|_{\tau=0}^{\text{war}} = 0. \tag{59}$$

Result 3.3 (War threshold is independent of N)**.** *In a symmetric trade war, every bystander's currency appreciates against both belligerents if and only if*

$$\boxed{\rho > \rho_1^* = \frac{\rho^*(N)}{N}} \tag{60}$$

for any number of countries. The belligerents' adjustments offset (e_{AB} is unchanged), so each bystander's equation decouples and its sign equals the sign of its own impact effect, a per-country condition that does not dilute with N. At $\eta = 1$, $\rho_1^* = 1 - \alpha_T(1-\alpha_D) < 1$, so any $\rho \geq 1$ produces joint depreciation of the belligerents; for $\eta > 1$, ρ_1^* can exceed one, and sufficiently low cross-origin *substitutability reverses even the war case.*

The formulas are verified against nonlinear equilibrium solves at $N = 4$ (threshold $\rho^*(4) = 5.28$ at $\alpha_D = 0.8$, $\alpha_T = 0.4$, $\eta = 1.5$, matching the numerical root to five decimal places).

3.3 Correspondence across N

The table collects each object and its analogue at every scale. $\rho_1^* = 1 + \alpha_D(\eta - 1) - \alpha_T(1 - \alpha_D)$ throughout.

object	$N = 2$	$N = 3$ (symmetric)	general N
stability = ML	$\varepsilon_X + \varepsilon_M > 1$ (21)	$D > 0$	J stable; $\det(-J) > 0$
sufficient primitive	gross substitutes (ε's ≥ 1)	$\eta \geq 1$	gross substitutes; $-J$ an M-matrix
sign rule	one impact, sign unconditional: appreciation	weights $\frac{1}{3}\left(\begin{smallmatrix}2 & 1\\ 1 & 2\end{smallmatrix}\right)$ on two impacts	sign $\left(\sum_j w_{ij} F_j\right)$, $w > 0$
isolated-tariff threshold	none (no bystander exists)	$\rho^* = 3\rho_1^*$	$N\rho_1^*$ (symmetric); roots of a degree-$(N-1)$ polynomial in general
war threshold	—	$\rho_1^* = \rho^*/3$	ρ_1^*, independent of N

4 Canonical PTA model: competing exporters

4.1 Setup

Countries A (importer), B and C (exporters). Two goods: x and numeraire y. Endowments: A holds only y; B and C hold x (and possibly y). Quasilinear preferences in each country,

$$U_i = u_i(x_i) + y_i, \qquad u_i' > 0,\ u_i'' < 0. \tag{61}$$

A levies specific tariffs t_B, t_C on imports of x from B and C. Let q_j be the price received by exporter j (the world price of its good) and p the domestic price of x in A. Both origins sell in A at the same domestic price:

$$p = q_B + t_B = q_C + t_C. \tag{62}$$

4.2 Demands and market clearing

From $u_i'(x_i) =$ (local price of x):

$$M_A(p) = x_A^d(p), \qquad E_j(q_j) = \omega_j - x_j^d(q_j), \quad j = B, C, \tag{63}$$

with $M_A' < 0$ and $E_j' > 0$. Market clearing for x:

$$M_A(p) = E_B(q_B) + E_C(q_C) = E_B(p - t_B) + E_C(p - t_C). \tag{64}$$

Equation (64) determines p given (t_B, t_C); then q_B, q_C follow from (62).

4.3 Comparative statics of a discriminatory tariff

Totally differentiate (64) with respect to t_B, holding t_C:

$$M'_A \frac{dp}{dt_B} = E'_B\Big(\frac{dp}{dt_B} - 1\Big) + E'_C \frac{dp}{dt_B} \quad \Longrightarrow \quad \frac{dp}{dt_B} = \frac{E'_B}{E'_B + E'_C - M'_A} \in (0,1). \tag{65}$$

Therefore

$$\frac{dq_B}{dt_B} = \frac{dp}{dt_B} - 1 = \frac{E'_C - M'_A}{E'_B + E'_C - M'_A} \cdot (-1) \;<\; 0, \tag{66}$$

$$\frac{dq_C}{dt_B} = \frac{dp}{dt_B} \;>\; 0. \tag{67}$$

Result 4.1 (Viner–Mundell)**.** *A discriminatory tariff on B worsens B's terms of trade and* im*proves C's terms of trade. Equivalently, a preferential tariff cut toward B (lower t_B) improves B's and worsens C's terms of trade.*

4.4 The PTA assumptions as restrictions on the three-country threshold

In the model of Section 2, an improvement in C's terms of trade corresponds to a rise in C's wage in common-currency units, that is, a rise in e_{AC}: A depreciates against C. The canonical PTA model delivers this sign unconditionally, from the slope conditions $E'_j > 0$ and $M'_A < 0$ alone, while Section 2 requires $\rho > \rho^*$. To see exactly why, decompose the threshold into its three claims on C's trade balance (values at the benchmark $\alpha_D = 0.8$, $\alpha_T = 0.4$, $\eta = 1.5$):

$$\rho^* = \underbrace{3}_{\substack{B\text{–}C \text{ linkage:} \\ B\text{'s income loss cuts its} \\ \text{purchases from } C;\ C \text{ switches} \\ \text{toward } B\text{'s cheapened goods}}} + \underbrace{3\alpha_D(\eta - 1)}_{\substack{\text{home switching:} \\ A\text{'s spending leaves} \\ \text{the import bundle}}} - \underbrace{3\alpha_T(1 - \alpha_D)}_{\substack{\text{openness credit:} \\ \text{more import spending} \\ \text{available to divert}}} = 3.0 + 1.2 - 0.24 = 3.96. \tag{68}$$

Each assumption of the canonical model removes one piece. Recomputing the threshold under each restriction (numerically, with the nested model of Section 2; severing trade changes the baseline, so the pieces are approximately rather than exactly additive):

restriction	threshold ρ^*	piece removed
none (benchmark)	3.96	—
no B–C trade ($\beta_{BC} = \beta_{CB} = 0$)	1.05	the linkage claim
no home tradables ($\alpha_D = 0$)	1.80	the home-switching claim
no non-tradables ($\alpha_T = 1$)	3.60	credit maximized (small alone)
$\alpha_D = 0$ and $\alpha_T = 1$	none: reversal for all $\rho > 0$	
no B–C trade and $\alpha_D = 0$	none: reversal for all $\rho > 0$	

Three readings of the table. First, the largest single piece of the threshold is the B–C linkage, not home switching: severing the target–bystander trade alone cuts the threshold from 3.96 to 1.05

even with full home bias and non-tradables. Second, non-tradables are not an independent opposing force: with the linkage severed and $\alpha_D = 0$, there is no threshold at any α_T; the non-tradable share matters only by shrinking the credit that offsets the other claims. Third, there are two routes to an unconditional sign — remove the home claim and maximize the credit ($\alpha_D = 0$, $\alpha_T = 1$), or remove the home claim and the linkage claim structurally — and the canonical model takes both at once, adds quasilinearity (which removes income effects wherever trade remains), and in the strict Viner case sets $\rho = \infty$.

Result 4.2. *The canonical PTA model is the restriction of the three-country model to a subspace of structures on which $\rho^* \leq 0$. The unconditional Viner–Mundell sign is the threshold condition of Section 2 evaluated where it cannot bind.*

The same reading organizes the literature. The competing-exporter models (Bagwell–Staiger 1999 and the GATT literature; Ornelas 2005) impose the severed topology and quasilinearity; the symmetric endowment-bloc models (Krugman 1991; Bond–Syropoulos 1996) impose full tradability with no distinct home-consumption margin and a single elasticity; the Viner tradition takes $\rho = \infty$; and Kemp–Wan 1976, which restricts none of these, proves existence and signs nothing. Applied models of preferential agreements (the GTAP-class CGE models, with a two-level Armington nest as standard practice, and modern quantitative trade models) keep all three pieces of (68) positive and generate configuration-dependent excluded-country signs — but report simulation output rather than the condition. The threshold of Section 2 is that condition.